

2026-08-04 Construction of the analysis sample for the BioPharm industry

Wang Wenzhi

August 4, 2026

Outline

- 1 The sample of focal new hires
- 2 The sample of candidate job postings
- 3 The pipeline of variable extraction from job postings

The workflow for constructing the sample of focal new hires

Start from users' profiles.

① Get a candidate sample of new hires.

- Retain employment spells in the BioPharm industries and scientist/engineer O*NET occupations. [Restrictions](#)
- Retain spells starting from January 2021 through December 2023. (Hopefully we can get at least two years of follow-up.)

② Apply sample restrictions.

- Exclude spells with missing required identifiers or an invalid/out-of-range start month.
- Exclude employment spells with missing country or state information.
- Exclude spells with neither a usable translated nor raw title.
- Exclude internship positions identified from the translated-title-first fallback.

③ Retain one focal spell within each user-company cell.

- Keep the earliest retained focal spell in the 2021-23 cohort window for each (user, company) cell.
- If there is a tie in the start year-month, then order employment spells within each (user, company) cell and keep only the first one.
 - smallest `seniority` value, with missing seniority ordered last;
 - smallest `position_number`, then smallest `position_id` as the final deterministic tie-break.

The description of the final sample of new hires

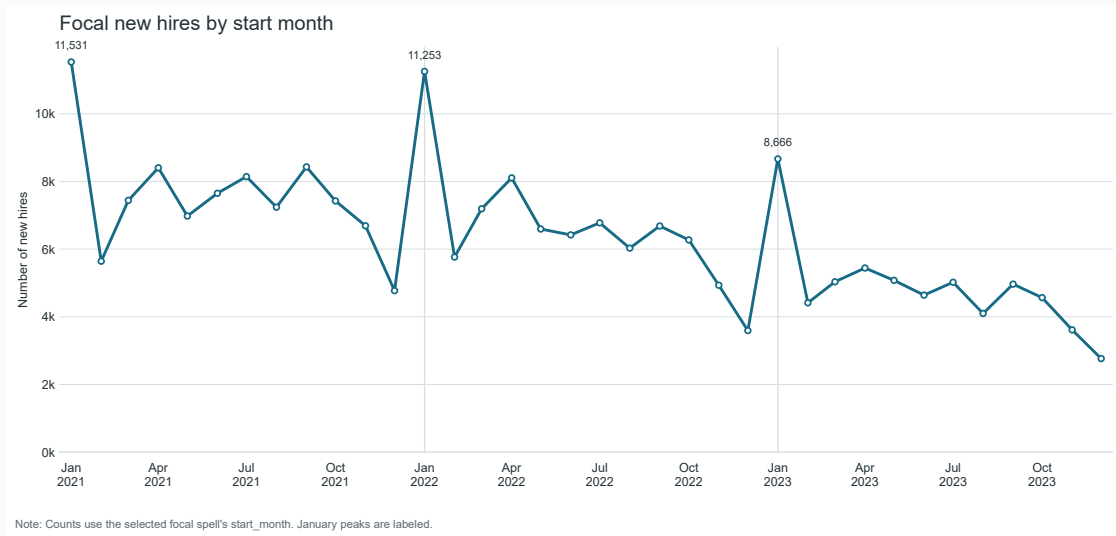
The final sample of new hires:

- It contains one row per user-company cell; the same user can appear at multiple companies.
- Each row is the earliest *retained focal-occupation spell* within that user-company cell during 2021–23.
- It is not necessarily the user's first spell at the company: internal transfers into a focal occupation are intentionally retained (e.g., if an accountant becomes a scientist within the same company, then I keep only his/her scientist employment spell and count as a new hire).
- Future-information flags are coverage diagnostics and do not restrict the baseline sample.

Sample size:

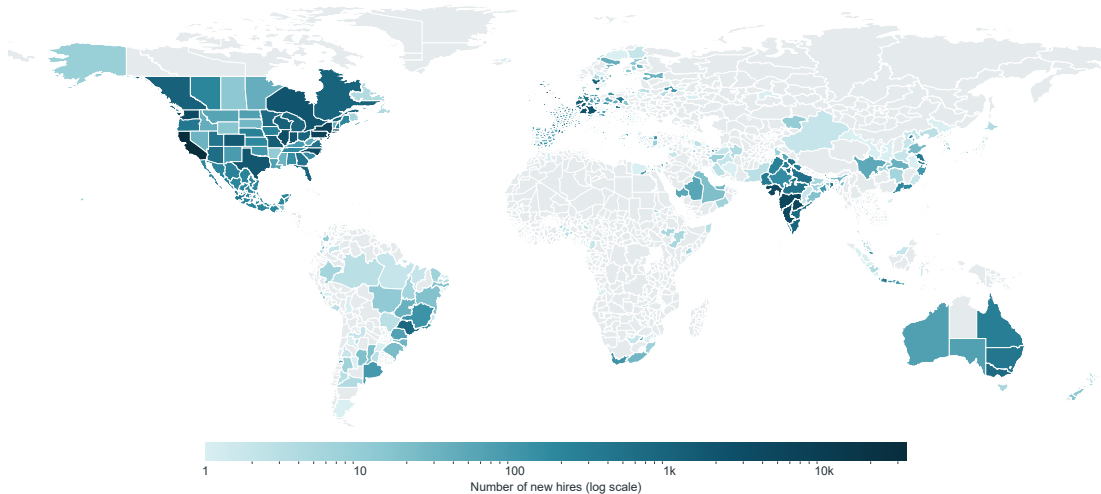
- After step 1, there are 294,915 candidate employment spells (239,957 unique users, 16,857 unique companies, 978 unique country-states).
- After step 2, there are 255,796 employment spells (208,988 unique users, 15,751 unique companies, 957 unique country-states).
- After step 3, there are 228,264 employment spells (208,988 unique users, 15,751 unique companies, 953 unique country-states).

Summary statistics of the final new hires sample: Time variation



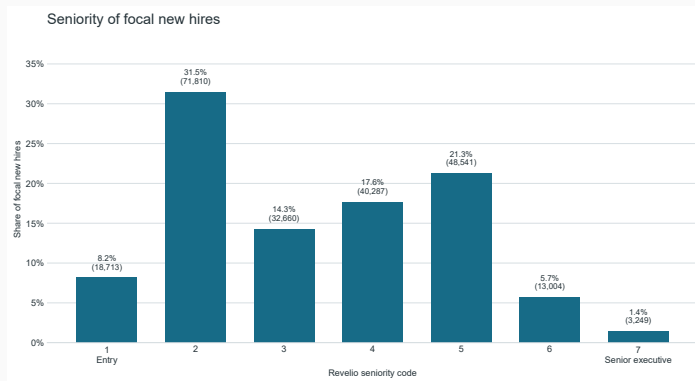
Summary statistics of the final new hires sample: Geographic variation

Focal new hires across country-states



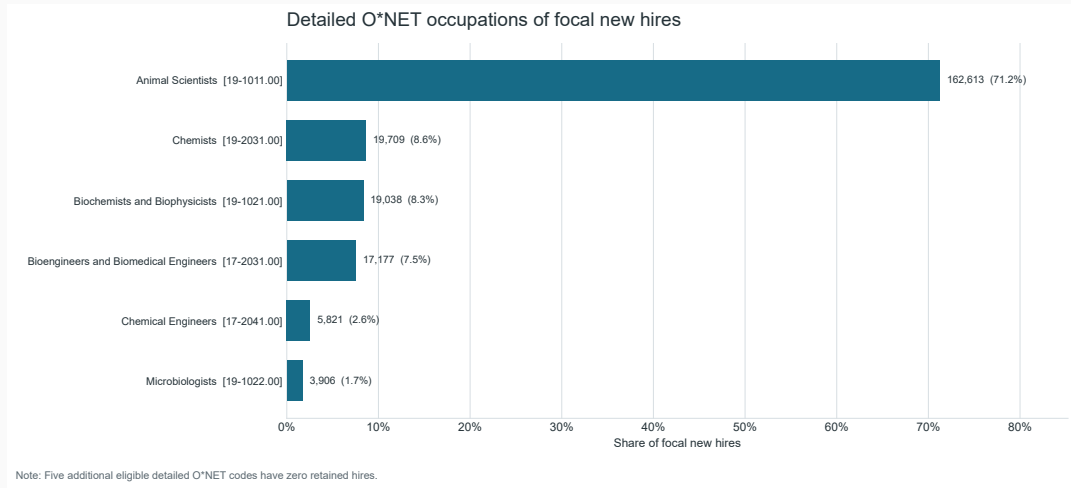
Largest country-states: California, United States: 34,173; Massachusetts, United States: 29,179; England, United Kingdom: 14,637; Pennsylvania, United States: 6,540; New York, United States: 6,172.
Mapped 227,164 of 228,264 hires (99.5%). Boundaries: Natural Earth 1:10m admin-1.

Summary statistics of the final new hires sample: Seniority distribution



- Around 30% of the new hires are at the junior level. [▶ The definition of seniority](#)
- Current sample construction process depends partially on the seniority of the position.
- Given the theme of our current project, junior hires may subject more to asymmetric information problem. But the within firm distribution of new hires' seniority may also reflect their talent discovery practices (hiring established scientists externally v.s. building up talent base internally).

Detailed O*NET occupations



Notes. There is 0 new hire for 5 pre-specified ONET occupations: "Bioinformatics Scientists", "Molecular and Cellular Biologists", "Geneticists", "Biologists", "Medical Scientists, Except Epidemiologists", "Bioengineers and Biomedical Engineers".

Flag new graduates and future-outcome observability

Final focal-hire sample: $N = 228,264$ user-company observations

New-graduate status

- A **first entrant** starts the focal job in the same month as the first substantive job in the complete observed history.
- A **new graduate** is a first entrant with a qualifying degree completed 12 months before through 3 months after the first job.
- I didn't spend too much time on cleaning the education dataset (really messy). I apply the most strict rule so that the new graduate subsample (though has a very small sample size) is the cleanest.

Status	Count	Share
New graduate	4,054	1.8%
Non-new graduate	216,058	94.7%
Unknown	8,152	3.6%

► Detailed procedures

Future-outcome observability

- Flag whether we can observe new hires' outcomes 12 and 24 months after the focal start.
- **Profile evidence**: latest `updated_dt`^a is on or after the checkpoint.
- **Later-position evidence**: maximum observed position start month is on or after the checkpoint.

Evidence	12 months	24 months
Profile updated	226,877 (99.4%)	187,331 (82.1%)
Later position	107,337 (47.0%)	60,698 (26.6%)

^aVariable label from Revelio: Date user's profile was last updated in our data.

Outline

- ① The sample of focal new hires
- ② **The sample of candidate job postings**
 - The messy job postings
- ③ The pipeline of variable extraction from job postings

The workflow for constructing the sample of candidate job postings

Inputs: The final sample of new hires and the raw LinkedIn postings table.

- ❶ **Construct one extraction window per focal company.** For each exact company identifier (`rcid`), extend from 18 months before its first focal hire through 3 months after its last focal hire.
- ❷ **Retain job postings within a narrow occupation of scientists/medical engineers.** Retain postings within the focal companies and one of the 11 focal O*NET codes. This yields 838,436 job postings.
- ❸ **Apply company-specific posting-date windows.** Retain inclusive boundary months based on the extraction window per focal company. This yields 742,963 job postings.
- ❹ **Exclude internship advertisements by title.** Normalize the translated title when available and otherwise the raw title; exclude only positive internship-title matches. Missing titles remain eligible. This removes 14,155 rows.

Comments:

- These are plausible postings for a company in the new hires sample, within a narrow occupation, and within the company's extraction window; they are not observed one-to-one matches between individual hires and advertisements.
- I can currently access to both LinkedIn and Indeed, but I focus on LinkedIn job postings only for now.
 - There is additional complexity when dealing with LinkedIn and Indeed in the same time.
 - Same job postings may appear twice in the two sources with distinct `job_id`.
 - I don't wish to design my own de-duplication algorithms, so I simply choose the source (LinkedIn) with more job postings.
- The sample of candidate LinkedIn job postings:
 - 728,808 job postings.
 - This seems to be substantially more job postings than the 228,264 focal hires, but it may not after removing some messy and duplicated job postings (More examples in the following).

Outline

- ① The sample of focal new hires
- ② The sample of candidate job postings
The messy job postings
- ③ The pipeline of variable extraction from job postings

A good and informative job posting

```
print(df[df["job_id"] == 2149708573]["description"].to_list())

# [As a Research Associate 2 within Illumina's IVD Assay Development, you will work as part of a multi-disciplinary team to support sample preparation in the generation of sequencing based assays for clinical oncology applications. In this role you will primarily be responsible for preparing and testing samples to support assay development and optimization.<br/><br/><strong><u>Tasks And Responsibilities</u></strong><ul><li>Plan and execute various laboratory activities with limited instruction</li><li>Perform complex experiments, including but not limited to: maintaining a detailed lab notebook, performing data analysis, and presenting results and updates for manager and teams as needed</li><li>Produce and test complex reagents</li><li>Communicate results and conclusions to aid in team decision making in a fast-paced research environment using a product development process</li><li>Communicate using various methods including email, presentations, and formal reports</li><li>Contribute to ongoing development activities, including various aspects of planning, feasibility, operation, and transfer</li><li>Manage goals and duties within defined target timeframes</li><li>Other duties may include instrument maintenance, reagent ordering and/or inventory, sample extractions, sample QC, and experiment troubleshooting.</br></li></ul>Listed responsibilities are an essential, but not exhaustive list, of the usual duties associated with the position. Changes to individual responsibilities may occur due to business needs.<br><br/><strong><u>Required Skills &amp; Experience</u></strong><ul><li>Bachelor's degree in biochemistry, molecular biology, genetics, genomics or related field.</li><li>3-5 years' experience in development of molecular biology and/or biochemical applications including DNA based techniques</li><li>Product development experience, including assay optimization, reagent development, integration, and testing, transfer to production and documentation</li><li>Hands-on experience with next generation sequencing technologies (NGS)</li><li>Ability to think and work independently as well as in a multidisciplinary team environment</li><li>Analytical, detail oriented, organized, efficient, problem solver with excellent verbal and written communication skills</li><li>Must have a desire to be part of a fast-paced and dynamic development environment</li><li>Must be fluent in use of Microsoft Word, Powerpoint, and Excel</li></ul>All listed requirements are deemed as essential functions to this position; however, business conditions may require reasonable accommodations for additional task and responsibilities.<br><br/><strong><u>Preferred Skills &amp; exp:</u></strong><ul><li>Experience with Illumina sequencing instruments and library preparation kits strongly preferred</li><li>Hands-on development or optimization of library preparation kits preferred</li><li>Experience with bioinformatics including data interpretation preferred</li><li>Hands on experience with FFPE and/or cDNA materials preferred</li><li>Knowledge of NGS secondary analysis techniques desirable</li><li>Training or experience in the field of oncology desirable</li><li>Clinical experience such as: IVD product development, Laboratory Developed Tests (LDT), and/or Companion Diagnostics (CDx) desirable</li></ul>At Illumina, we strive to foster a diverse and inclusive workplace by cultivating an environment in which everyone contributes to our mission. Built on a strong foundation, Illumina has always been rooted in openness, collaboration, and seeking alternative views and perspectives to propel innovation in genomics. We are proud to confirm a zero-net gap in pay, regardless of gender, ethnicity, or race. We also have several Employee Resource Groups (ERG) that deliver career development experiences, increase cultural awareness, and demonstrate our collective commitment to diversity and inclusion in the communities we live and work. We are proud to be an equal opportunity employer committed to providing employment opportunity regardless of sex, race, creed, color, gender, religion, marital status, age, national origin or ancestry, physical or mental disability, medical condition, sexual orientation, pregnancy, military or veteran status, citizenship status, and genetic information. If you require accommodation to complete the application or interview process, please contact accommodations@illumina.com. To learn more, visit: https://www.dol.gov/ofccp/regs/compliance/posters/pdf/eeopost.pdf']
```

- HTML tags to indicate the structure of the job posting.
- Relevant sections: "Tasks and Responsibilities", "Required Skills & Experience", "Preferred Skills & Experience", some general introduction of the company in the end.

Bad job postings: Empty text

```
# Null description and exact source sentinels.
print(df[df["job_id"] == 132367778]["description"].to_list())
# [None]

print(df[df["job_id"] == 2149963273]["description"].to_list())
# ['N/A']

print(df[df["job_id"] == 2369898471]["description"].to_list())
# ['None']

# Markup shell with no visible text.
print(df[df["job_id"] == 2517768054]["description"].to_list())

# Punctuation, symbols, or invisible Unicode only.
print(df[df["job_id"] == 2162308930]["description"].to_list())
# ['.', '.']

print(df[df["job_id"] == 2175458444]["description"].to_list())
# ['', '']

print(df[df["job_id"] == 2322300942]["description"].to_list())
# ['\u200b']
```

Bad job postings: No actual contents

```
# Employer, ATS, or social-media URL only.
print(df[df["job_id"] == 2151163725]["description"].to_list())
# ['www.biontech.de']

print(df[df["job_id"] == 2149693498]["description"].to_list())
# ['https://j.brt.mv/ATS/PortalViewRequirement.do?reqGK=27415250']

print(df[df["job_id"] == 2150975193]["description"].to_list())
# ['https://www.linkedin.com/company/abalone-bio-inc']

# Email address only.
print(df[df["job_id"] == 2182318787]["description"].to_list())
# ['careers@redpinrx.com']

print(df[df["job_id"] == 2208421400]["description"].to_list())
# ['E-mail: bonsdorff@oncoinvent.com']

print(df[df["job_id"] == 2311789900]["description"].to_list())

# Date only.
print(df[df["job_id"] == 2169425610]["description"].to_list())
# ['17th January 2021']

print(df[df["job_id"] == 3487258026]["description"].to_list())
# ['March 2023']

print(df[df["job_id"] == 2959422594]["description"].to_list())
# ['March 22, 2022']

# Agency or recruiter notice only.
print(df[df["job_id"] == 2151158203]["description"].to_list())
# ['No Agencies or Recruiters']

print(df[df["job_id"] == 2159765846]["description"].to_list())
# ['No recruiters please']

print(df[df["job_id"] == 2813519909]["description"].to_list())
# ['Principals only. No agencies please.']
```

Bad job postings: No actual contents

```
# Application-document or application-redirect fragment.
print(df[df["job_id"] == 2465425031]["description"].to_list())
# ['To apply, please email your CV and cover letter to careers@biofidelity.com.']

print(df[df["job_id"] == 2445786979]["description"].to_list())
# ['Send CV and Cover letter to admin@limmtx.com']

print(df[df["job_id"] == 3272096610]["description"].to_list())
# ['Apply at www.evolveimmune.com']

# Location or job-level fragment.
print(df[df["job_id"] == 2149785298]["description"].to_list())
# ['\xa0\xa0\xa0\xa0\xa0Level will be commensurate with experience']

print(df[df["job_id"] == 2166953970]["description"].to_list())
# ['• Salt Lake City Metro Area, UT']
```

Bad job postings: Fragments of full job postings

```
# Compensation fragments.
print(df[df["job_id"] == 3524211563]["description"].to_list())
# ['salary: range (50k - 75k)']

print(df[df["job_id"] == 3607273029]["description"].to_list())
# ['Salary: Range: $65,000 - $95,000']

print(df[df["job_id"] == 3836579571]["description"].to_list())

# Vaccination or employment-condition fragments.
print(df[df["job_id"] == 3343763871000000000004]["description"].to_list())
# ['COVID-19 vaccine is required.']

print(df[df["job_id"] == 2861019658]["description"].to_list())

print(df[df["job_id"] == 2890360978]["description"].to_list())

# Isolated education or experience requirements.
print(df[df["job_id"] == 3605667513]["description"].to_list())
# ['PhD - 1-4 Years of experience']

print(df[df["job_id"] == 2299511922]["description"].to_list())
# ['Bachelor's degree and 7 years of Scientific experience']

print(df[df["job_id"] == 3294469443]["description"].to_list())

# Isolated skills or responsibility statements.
print(df[df["job_id"] == 2513185360]["description"].to_list())
# ['HPLC, GC']

print(df[df["job_id"] == 2481383658]["description"].to_list())
# ['Flow cytometry']

print(df[df["job_id"] == 3306839629]["description"].to_list())
# ['Familiarity with NGS workflows']
```

Bad job postings: Templates only without actual job-specific contents

```
# Legal or equal-employment-opportunity boilerplate only.
print(df[df["job_id"] == 35202486690000000002]["description"].to_list())
# ['Equal Opportunity Employer:']

print(df[df["job_id"] == 2148292480]["description"].to_list())
# ['Equal Opportunity Employer: All qualified applicants will receive consideration for employment without regard to race, color, religion, sex, sexual orientation, gender identity, national origin, disability, or status as a protected veteran.']

print(df[df["job_id"] == 2148143737]["description"].to_list())
# ['PACT Pharma is proud to be an equal opportunity employer and strives to build a diverse and inclusive team. We do not discriminate on the basis of race, color, national origin, religion, gender, sexual orientation, age, marital status, veteran status, or disability status.']

# Applicant privacy or data-processing notice only.
print(df[df["job_id"] == 29948316590000000003]["description"].to_list())
# ['See our Privacy Policy.']

print(df[df["job_id"] == 2438181917]["description"].to_list())
# ['Please review our Job Applicant Privacy Policy at http://www.paragongenomics.com/job-privacy-policy/ for information about how your personal data is processed and stored.']

print(df[df["job_id"] == 2583039984]["description"].to_list())
# ['Genmab A/S is committed to protecting your personal data and privacy. Please see our privacy policy for handling your data in connection with your application on our website at https://www.genmab.com/privacy.']

# Company website, profile, or social-media redirect only.
print(df[df["job_id"] == 2805182341]["description"].to_list())
# ['Visit our website: https://www.sixfold.bio/']

print(df[df["job_id"] == 2311407679]["description"].to_list())
# ['For up-to-date information about Apotex's ongoing efforts in response to COVID-19, visit: https://www1.apotex.com/global/covid-19-update or follow us on LinkedIn and Twitter']

print(df[df["job_id"] == 2181340319]["description"].to_list())
# ['Ab Studio Inc. is a well-funded start-up Biotech located in San Francisco bay area, USA. Founder and leading scientists have extensive expertise in the field of therapeutic antibody discovery, engineering, humanization and optimization. As a start-up biotech, Ab Studio aims to support the whole therapeutic antibody R&D field with 1) novel technology platforms on making bispecific and internalization antibodies 2) high quality antibody design and engineering services and partnership. We deeply believe in the "quality by design" concept and wish our work can make significant contributions to the whole therapeutic antibody development field. Please visit our website: www.antibodystudio.com for more details.']
```

Bad job postings: Evergreen job postings

```
print(Evergreen or talent-post posting that is not a current opening.  
{"id":"dfj0f job id" : 2988288654["description"] ,to_list()})  
# I 'sourcing Statement\nI am applying to be part of the Novartis Future oppo. We are not  
currently recruiting for this role but we are building a pipeline for future opportunities. If you  
are interested please email me at [email address] or visit our website www.novartis.com/VW-NOVA  
Description\nThe Next Generation Scientist Program is an intensive training program for talented  
and motivated research scientists from low- and middle-income countries. The 8-week program is  
delivered virtually and consists of virtual classroom sessions, group work assignments and active  
individual case studies. Participants will receive both scientific and soft skills training and  
professional development of the post-graduate students. Guided by Novartis mentors, the  
participants learn about the pharmaceutical drug discovery and development process as well as  
participate in a leadership development program designed to enhance decision-making and  
communication skills. Upon successful completion, you will jointly receive a certificate from Novartis.  
Scientists with background and interest in malaria, sickle cell disease, leprosy or Chagas disease  
are encouraged to apply.\n\nApplication:\nPlease make sure to add the following documents to  
your application, combined in one PDF file: CV or resume letter of recommendation/support from  
a faculty member letter of motivation how the program will benefit you\n\nMinimum Requirements:\nEnrolled at university as PhD or MSc student in a geography  
of scope (low- and middle-income country) Research background in malaria, sickle cell disease,  
leprosy or Chagas disease. Good command of English. Willingness to participate in oral and  
written English communication skills Ability to work independently and in groups Good  
organizational skills Solid internet infrastructure for online training Language:  
English\n\nDuration and dates: 8-week program from July to August 2022; three weekly classroom  
sessions (Monday each)\nPreferred field of studies: Natural and Pharmaceutical Sciences,  
Engineering and Technology, Life Sciences, Energy and Environment That's how many lives our  
products touch in 2019. And while we're proud of that, in this world of digital and  
technological transformation, we must also ask ourselves this: how can we continue to improve and  
extend even more people's lives? In what ways do we believe the answers are found when curious, courageous and  
collaborative minds come together? At Novartis, we know the answer lies in bold decisions and taking  
smarter risks. Now are Novartis. Join us and help reshape medicine.
```

[illegible]

- I currently exclude evergreen job postings from the analysis sample.

Bad job postings: Non English

```
#・Non-English advertisement, including a case with an English legal footer.  
print(df[df["job_id"].isin([2843094717000000000000])["description"].to_list())  
#・['この職位は来源于招聘<br/><br/>职责描述:<br/><br/>儿童内分泌领域机制机理研究;<br/><br/>任职要求:<br/><ul><li>  
博士<br/></li></ul>2.2年以上代谢领域研究经验']  
  
print(df[df["job_id"].isin([3651420767])["description"].to_list())  
#・['<strong>職務内容:<br/><br/></strong>医薬品の製造業務<br/><br/>医薬品製造に関する必要な書類の記載、作成、改訂業務<br/><br/>作業時間や出来高量などシステムへの簡単なデータ入力業務<br/><br/>派遣スタッフへの業務指導<br/><br/>機械のキャリブレーション業務、メンテナンス立ち合い<br/><br/>改善提案の提出<br/><br/>会社が実施するイベントへの参加<br/><br/>など<br/><br/>仕事の魅力:<br/><br/>「ものづくり」をしながら、製品として形になる達成感とチームメンバーとの協調性を日々実感し、楽しく仕事をする事ができる。」<br/><br/>At Sanofi diversity and inclusion is foundational to how we operate and embedded in our Core Values. We recognize to truly tap into the richness diversity brings we must lead with inclusion and have a workplace where those differences can thrive and be leveraged to empower the lives of our colleagues, patients and customers. We respect and celebrate the diversity of our people, their backgrounds and experiences and provide equal opportunity for all.']
```

- Non-English advertisements themselves are not a problem.
- They make the design of the rule-based algorithms and human validation (much) more difficult.

Bad job postings: Lost sentence boundaries

```
# Lost sentence or section boundaries in run-on scraped text.
print(df[df["job_id"] == 28179205390000000003]["description"].to_list())
# ['Key Responsibilities\nContribute ideas to technology development strategies and maintain a culture of continual improvement and scientific excellencePerform conjugation of small molecules (e.g. cytotoxic drugs) to antibodiesPerform purification of antibody and ADCs by column chromatography (HIC, SEC, membrane ultrafiltration (TFF))Analyse the conjugates using analytical chromatography techniques (HIC, SEC), UV-Vis spectrophotometry, FACS and LC-MSPerform the interpretation of analytical resultsOperate process and analytical equipment and ensure its maintenancePresent results in written reports and oral presentations at team meetings']

print(df[df["job_id"] == 2681088216]["description"].to_list())
# ['Establish, maintain and differentiate iPSCs and primary cells to desired cell types.Perform CRISPR-based gene-editing and construction of stable cell lines in iPSC lines to drive target ID/validation and assay development across multiple therapeutic programs.Develop and optimize quantitative in vitro efficacy assays in iPSCs and/or primary cells using a variety of techniques (genetic engineering, high content imaging, FACS, HTRF, luminescence, etc).Actively participate on a dynamic and collaborative team focused on developing new medicines for patients in need.Independently interpret data and recommend next steps.Contribute to a fun, intellectually stimulating and fast-paced work environment.']

print(df[df["job_id"] == 2586684052]["description"].to_list())
# ['About TrueBinding, Inc.TrueBinding, Inc. is a biotherapeutics development company focused on creating new and exciting antibody for metabolic diseases and cancer, based on our proprietary TrueBinding platform with great unmet medical needs.JOB SUMMARYWe are seeking a highly motivated structural biologist to study protein complexes discovered by Truebinding team.Preferred CandidatesPhD/Master with focus on structural biologyExperience with large protein complex researchKnowledge of state-of-the-art protein structural biology techniques such as crystallization, structure determination and analysis.Experience in single particle cryo-EM workflow including grid freezing and screening, data collection and processing, three-dimensional reconstruction and model building.']
```

- This may not be a huge issue.

Takeaway from these bad examples

- The sample size of focal new hires is $N = 228,264$, of all candidate job postings is $N = 728,808$.
 - First, the sample of final job postings shouldn't consist of those bad job postings listed as above.
 - The construction of the job postings sample should be combined with the textual analysis part of the project, which will be introduced later.
 - Second, we shouldn't weight each job posting as equally important.
 - I understand that we don't need to achieve a one-to-one match from new hires to job postings.
 - However, it is not reasonable to treat all job postings as equally important.
 - Ideally, we still need to match new hires to a smaller subset of job postings (based on detailed job titles, geography information, posting dates, etc) and then assign weights to this smaller set of job postings.
 - The important practical problem is how to determine employee-specific subset of job postings, and how to assign weights.
 - We should interpret the weighted sample of job postings as the **average degree of selectivity a new hire faces for similar jobs within the company**.

Outline

- ① The sample of focal new hires
- ② The sample of candidate job postings
- ③ The pipeline of variable extraction from job postings
 - Overview of the 4 stages
 - Stage A. Pre-process job postings

Overview of the 4 stages

- Stage A: Pre-process job postings
 - Clean HTML tags while retaining document structure.
 - Identify informative job postings which will be sent to stage B for LLM variable extraction (more on this point later).
 - **Human validation is necessary to establish a reasonable and robust rule to identify informative job postings.**
- Stage B: Extract variables with the LLM
 - Focused modules return schema-constrained JSON claims with exact evidence.
 - **This is the only black-box stage:** LLM's internal reasoning is not observable.
 - **An important difference from other papers using textual algorithms:**
 - We are not trying to measure some abstract concepts like trust or policy uncertainty from the texts.
 - What we want are variables that have correct values (the minimum requirements for education or working experiences; the required technical skills).
- Stage C: Validate and normalize LLM outputs
 - LLM will output a series of json files linking extracted variables to the corresponding evidence in the texts.
 - Human validation is necessary to adjust how LLM works to construct variables, e.g., prompts, different models, etc.
- Stages A, C, and D should be deterministic, explicit, rule-based algorithms. The only 'stochastic' part is stage B where we use LLM to interpret job advertisements texts and construct variables.

Outline

- ① The sample of focal new hires
- ② The sample of candidate job postings
- ③ The pipeline of variable extraction from job postings
 - Overview of the 4 stages
 - Stage A. Pre-process job postings

Stage A: Pre-process job postings and identify informative posting texts

The aim of stage A:

- Stage A defines the job-posting text sample that can enter the LLM extraction stage. It does not measure job selectivity itself.
- The aim of this stage is to send a job description to Stage B only when the automated evidence indicates that **it is a sufficiently complete, current or unspecified, single-role, English job advertisement.**

Some guidelines:

- Process repeated descriptions once, but map every decision and later LLM result back to each source job posting.
- Preserve the original wording and provenance; normalization will not remove negation, alternatives, comparison signs, or other economically meaningful text.
- My current practice is to retain English job postings only (for the ease of validation).
- Partial fragments, uncertain language, explicit noncurrent/evergreen notices, and multi-role advertisements are excluded rather than guessed at by the LLM.

728,808 posting rows → 267,408 unique raw texts → 241,971 unique normalized texts

Step 1 of stage A. Preserve and parse raw job postings

Preserve source identity and avoid repeated work

- Keep one map row for every `job_id` and classify null, empty, whitespace-only, sentinel, or usable raw descriptions.
- Assign each exact non-null string a SHA-256 fingerprint and retain its posting multiplicity.
- De-duplicate only for computation: identical descriptions are parsed and later submitted to the LLM once, without dropping postings.

Recover document structure without rewriting meaning

- Remove noncontent HTML elements (e.g., scripts and styles) and parse headings, paragraphs, list items, table rows, and plain-text lines.
- Standardize Unicode, spacing, line endings, and HTML entities while preserving the visible wording and punctuation.
- Create an ordered canonical block representation, a normalized-text fingerprint, and a raw-to-normalized crosswalk.

Step 2 of stage A. Develop a rule to identify informative job advertisements texts

Evidence used by this step

- Length and structure: characters, words, headings, paragraphs, lists, and informative units.
- Text quality: parser warnings, replacement characters, run-on text, unmatched delimiters, possible truncation, URLs, and emails.
- Content: role/context, duties/responsibilities, candidate requirements, application/employment information, and boilerplate.
- Offline document- and block-level language scores classify each text as English, non-English, mixed, or uncertain.

Current rules to identify informative postings

Diagnostic	Required value
Visible characters	≥ 600
Words	≥ 100
Informative units ^a	≥ 3
Substantive domains ^b	≥ 2
Boilerplate share ^c	$\leq 80\%$
English score ^d	≥ 0.90
English margin ^d	≥ 0.20

^a A normalized heading, list item, or paragraph with at least 20 alphanumeric characters counts as one unit; a paragraph longer than 300 characters instead contributes its sentence-boundary count (at least one).

^b Number of distinct matched domains among role/context, duties/responsibilities, candidate requirements, and application/employment; duties or requirements must be present.

^c Characters in blocks that match a boilerplate phrase but no substantive domain, divided by characters in all blocks.

^d Offline Lingua score (calculated from a Python package); margin is the English score minus the strongest non-English score. English must also rank first, have at most 10% non-Latin letters, and show no block-level mixed-language disagreement.

Step 3 of stage A. Human validation (not conducted)

Human validation with a random sample

- I sample around 1,600 unique normalized texts.
- In principle, I (or other human reviewers) should label whether each text whether it is a full advertisement, English, current/unspecified, single-role, and of usable text quality.
- And then human feedbacks will be used to revise current rules so that we can keep the informative job postings for stage B LLM variable extraction.
- I don't have enough time to do this validation step. So my current approach is more like a proof of concept.

Attrition under current rules

Status	Unique texts	Postings
Eligible full English	118,895	367,019
Partial/incomplete	94,506	260,131
Non-English	21,810	78,503
Language uncertain	6,267	19,215
Other no-call	493	3,940
Total	241,971	728,808

Under current rules to identify informative job postings,

- Eligible shares are $49.1\% = 118,895/241,971$ of unique normalized texts and $50.4\% = 367019/728808$ of source postings.
- The difference reflects the multiplicity of repeated descriptions. Also, source postings without a normalized text appear only in the posting count.

A real example after Stage A: raw source

Raw description excerpt (HTML as stored)

```
<strong>We are seeking an experienced and motivated individual to join our team and expand our high resolution single B-cell screening platform using the Berkeley Lights Beacon.  ... </strong>
<p><br/></p>
<strong><u>Primary Responsibilities</u></strong>
<ul>
  <li>Application of microfluidics-based selections of antigen-specific B cells using the Berkeley Lights Beacon.</li>
  <li>Detailed reagent preparation and validation</li>
  <li>Design, implement, troubleshoot and validate antibody-based functional assays ...</li>
</ul>
<strong><u>Qualifications</u></strong>
<ul>
  <li>Advanced degree in biological sciences with at least 4 years of experience in a laboratory setting ...</li>
  ...
</ul>
```

Notes. This is an actual Stage A input. Ellipses shorten the display only; the complete raw strings remain unchanged in the raw-text table.

A real example: eligible normalized text

Role context and responsibilities

[PARAGRAPH] We are seeking an experienced and motivated individual to join our team and expand our high resolution single B-cell screening platform using the Berkeley Lights Beacon. ...

[HEADING] Primary Responsibilities

- [LIST ITEM] Application of microfluidics-based selections of antigen-specific B cells using the Berkeley Lights Beacon.
- [LIST ITEM] Detailed reagent preparation and validation
- [LIST ITEM] Design, implement, troubleshoot and validate antibody-based functional assays (protein-protein interactions and cell-based), particularly on the nanoscale level.
-
-
-

Qualifications

[HEADING] Qualifications

- [LIST ITEM] Advanced degree in biological sciences with at least 4 years of experience in a laboratory setting (strong preference to industry experience).
- [LIST ITEM] Ideal candidates will have prior experience in single B-cell (i.e., B cell cloning)-based antibody discovery workflows ...
- [LIST ITEM] Prior experience establishing and validating antibody-based functional assays and characterization processes.
- [LIST ITEM] Exceptional verbal and written communication skills.
-
-
-

A real example after Stage A: why it passes

Assessment of eligibility under current rules

Diagnostic	Actual	Gate	Pass
Visible characters	1,986	≥ 600	Yes
Words	252	≥ 100	Yes
Informative units	17	≥ 3	Yes
Substantive domains	3	≥ 2	Yes
English score	1.00	≥ 0.90	Yes
English margin	1.00	≥ 0.20	Yes
Boilerplate share	0.0%	$\leq 80\%$	Yes
Parser warnings	0	none	Yes
Corruption / truncation	No	none	Yes

My current progress and next steps

- I have basically finished stage A pipeline to identify those informative job postings so that they can be sent to stage B for LLM to construct variables.
 - However, current stage A rules are only temporary because human validation is necessary to assess if current rules are reasonable enough.
 - For example, is the restriction “Visible characters ≥ 600 ” too strict so that we lose many informative job postings?
- Step B LLM variable construction requires us to use LLM API keys.

Industry and ONET occupation restrictions

► Back

- Industries are restricted to (Revelio standardized variable `rics_k400`)
 - "Biotechnology and Life Sciences"; "Pharmaceutical Manufacturing"; "Pharmaceuticals"
- ONET occupations are restricted to (Revelio predicted ONET occupation `onet_code`)
 - "19-1011.00": "Animal Scientists"
 - "19-1021.00": "Biochemists and Biophysicists"
 - "19-1022.00": "Microbiologists"
 - "19-1029.01": "Bioinformatics Scientists"
 - "19-1029.02": "Molecular and Cellular Biologists"
 - "19-1029.03": "Geneticists"
 - "19-1029.04": "Biologists"
 - "19-2031.00": "Chemists"
 - "19-1042.00": "Medical Scientists, Except Epidemiologists"
 - "17-2031.00": "Bioengineers and Biomedical Engineers"
 - "17-2041.00": "Chemical Engineers"

The definition of seniority

► [Back to the tenure distribution](#)

- The seniority metric is created using an ensemble model.
 - First, information about an individual's current job, including their title, company, and industry, are used to generate an initial seniority score.
 - Second, details about an individual's job history, such as the duration of their previous employment and the seniority of previous positions, are taken into account to create a second seniority score.
 - Finally, an individual's age is used to generate a third seniority score. The scores from these models are averaged together to arrive at a continuous seniority metric for an individual.
- The seven ordinal seniority levels are:
 - Entry Level (Ex. Accounting Intern, Software Engineer Trainee, Paralegal)
 - Junior Level (Ex. Account Receivable Bookkeeper, Junior Software QA Engineer, Legal Adviser)
 - Associate Level (Ex. Senior Tax Accountant; Lead Electrical Engineer; Attorney)
 - Manager Level (Ex. Account Manager; Superintendent Engineer; Lead Lawyer)
 - Director Level (Ex. Chief of Accountants; VP Network Engineering; Head of Legal)
 - Executive Level (Ex. Managing Director, Treasury; Director of Engineering, Backend Systems; Attorney, Partner)
 - Senior Executive Level (Ex. CFO; COO; CEO)

Current procedures to identify new graduates

[▶ Back to the sample construction](#)

Classification is performed at the **user level** using complete observed position and education histories, then attached to every focal user–company observation.

1. First observed substantive job

- A candidate record must have a valid start month and a usable raw or translated title.
- Exclude titles identified as internships or placements.
- Treat research, teaching, or graduate assistants; student or undergraduate researchers; and doctoral/PhD students or candidates as education-linked candidates.
- Exclude such a candidate only when its position interval overlaps a valid education interval. Open-ended positions are treated as continuing; postdoctoral and generic fellow roles remain jobs.
- The first substantive job month is the earliest remaining start month across the user’s full history.

Employment-history safeguard

Mark the result unreliable if there is no usable first job, any record lacks a start month, position ID, position number, or title, or the minimum observed position number exceeds one. These cases are classified as *unknown* before education is considered.

2. Qualifying degree completion

- Normalize degree text; use the standardized field when it maps uniquely, otherwise use the raw field.
- Qualifying levels are bachelor’s, master’s, and research doctorates (PhD, DPhil, ScD, or DSc). Certificates, secondary/associate education, and named professional doctorates such as MD, JD, and PharmD do not qualify; ambiguous text remains unknown.
- Require a usable, non-reversed completion month and $-3 \leq \text{first-job month} - \text{degree-end month} \leq 12$. Thus, graduation may occur up to 12 months before or 3 months after entry.
- If several degrees qualify, select the closest completion, then the highest degree level, then the largest nonmissing `education_number`.
- Education evidence is insufficient with no education record, a key conflict, any unknown degree level, or a qualifying degree with an unusable end month.

3. Classification order (first matching rule)

Unreliable employment history	Unknown
Focal start is after first substantive job	Non-new
Focal start equals first job; qualifying degree	New graduate
Focal start equals first job; education insufficient	Unknown
Focal start equals first job; adequate evidence, no qualifying degree	Non-new